

Review of Some Topics Covered in the Course

The Fundamental Theorem of Calculus - Part 1 (FTC-1)

If f is continuous on $[a, b]$, then the function g given by

$$g(x) = \int_a^x f(t) dt \quad (a \leq x \leq b)$$

is differentiable on $[a, b]$ and $g'(x) = f(x)$ for all $x \in [a, b]$.

The Fundamental Theorem of Calculus - Part 2 (FTC-2)

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = \left[F(x) \right]_a^b = F(b) - F(a)$$

Suggested practice questions:

1. Examples 5 and 6 of Section 5.3 in the lecture outline.
2. Related questions from Section 5.3 in Homework 2.

The Substitution Rule If $f(x)$ is continuous and $u = g(x)$ is differentiable, then

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

Since $u = g(x)$, $du = g'(x) dx$. So, the Substitution Rule can be formally obtained by replacing $g(x)$ by u and replacing $g'(x) dx$ by du . This rule has the following special case:

The General Power Rule (GPR) If $u(x)$ is differentiable, then

$$\int u(x)^n u'(x) dx = \frac{1}{n+1} u(x)^{n+1} + C \quad (n \neq -1), \quad \int \frac{u'(x)}{u(x)} dx = \ln |u(x)| + C$$

Suggested practice questions:

1. Examples 8, 9, 12 and 13 of Section 5.5 in the lecture outline.
2. Related questions from Section 5.5 in Homework 2.

Suppose f' and g' are continuous functions.

Integration by Parts for Indefinite Integral

$$\int f(x) g'(x) dx = f(x) g(x) - \int f'(x) g(x) dx$$

Integration by Parts for Definite Integral

$$\int_a^b f(x) g'(x) dx = \left[f(x) g(x) \right]_a^b - \int_a^b f'(x) g(x) dx$$

Suggested practice questions:

1. Example 1 of Section 7.1 in the lecture outline.
2. Related questions from Section 7.1 in Homework 3.

I. Integral of the type $\int \sin^m x \cos^n x dx$ ($m, n \geq 0$ are integers)

(a) $\int \sin^{2k+1} x \cos^n x dx = \int (1 - \cos^2 x)^k \cos^n x \sin x dx = - \int (1 - u^2)^k u^n du$ with $u = \cos x$

(b) $\int \sin^m x \cos^{2k+1} x dx = \int \sin^m x (1 - \sin^2 x)^k \cos x dx = \int u^m (1 - u^2)^k du$ with $u = \sin x$

(c) $\int \sin^{2k} x \cos^{2\ell} x dx = \int \left(\frac{1}{2}(1 - \cos 2x)\right)^k \left(\frac{1}{2}(1 + \cos 2x)\right)^\ell dx$ (reduce the power)

II. Integral of the type $\int \tan^m x \sec^n x dx$ ($m, n \geq 0$ are integers)

(a) $\int \tan^{2k+1} x \sec^n x dx = \int (\sec^2 x - 1)^k \sec^{n-1} x (\sec x \tan x) dx = \int (u^2 - 1)^k u^{n-1} du$

with $u = \sec x$

(b) $\int \tan^m x \sec^{2k} x dx = \int \tan^m x \sec^{2k-2} x \sec^2 x dx = \int u^m (1 + u^2)^{k-1} du$ with $u = \tan x$

Suggested practice questions:

1. Examples 2 - 5 and 8 of Section 7.2 in the lecture outline.
2. Related questions from Section 7.2 in Homework 3.

I. Trigonometric Substitution for $\sqrt{a^2 - x^2}$

Let $x = a \sin \theta$. Then $dx = a \cos \theta d\theta$, $x^2 = a^2 \sin^2 \theta$ and $\sqrt{a^2 - x^2} = a \cos \theta$.

II. Trigonometric Substitution for $\sqrt{x^2 + a^2}$

Let $x = a \tan \theta$. Then $dx = a \sec^2 \theta d\theta$, $x^2 = a^2 \tan^2 \theta$ and $\sqrt{x^2 + a^2} = a \sec \theta$.

III. Trigonometric Substitution for $\sqrt{x^2 - a^2}$

Let $x = a \sec \theta$. Then $dx = a \sec \theta \tan \theta d\theta$, $x^2 = a^2 \sec^2 \theta$ and $\sqrt{x^2 - a^2} = a \tan \theta$.

Suggested practice questions:

1. Examples 9 - 12 of Section 7.3 in the lecture outline.
2. Related questions from Section 7.3 in Homework 3.

Integration of Rational Functions by Partial Fractions

Let $\frac{R(x)}{Q(x)}$ be a proper rational function.

(1) If $Q(x) = (a_1x + b_1)^{r_1} \cdots (a_kx + b_k)^{r_k}$ with $r_1, \dots, r_k \geq 1$, then $\frac{R(x)}{Q(x)}$ has the partial fraction decomposition

$$\frac{R(x)}{Q(x)} = \left(\frac{A_{1,1}}{a_1x + b_1} + \cdots + \frac{A_{1,r_1}}{(a_1x + b_1)^{r_1}} \right) + \cdots + \left(\frac{A_{k,1}}{a_kx + b_k} + \cdots + \frac{A_{k,r_k}}{(a_kx + b_k)^{r_k}} \right)$$

(2) If $Q(x)$ contains an irreducible quadratic form $ax^2 + bx + c$ with $b^2 - 4ac < 0$, then the partial fraction decomposition of $\frac{R(x)}{Q(x)}$ contains the term $\frac{Ax + B}{ax^2 + bx + c}$.

Suggested practice questions:

1. Examples 13 - 15 of Section 7.4 in the lecture outline.
2. Related questions from Section 7.4 in Homework 4.

Improper Integrals over Infinite Intervals (a, ∞) .

We define $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$ if the limit exists.

Improper Integrals with Discontinuous Integrands.

(1) If f is continuous on $(a, b]$ and discontinuous at a , we define

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx \quad \text{if the limit exists.}$$

(2) If f is continuous on $[a, b)$ and discontinuous at b , we define

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx \quad \text{if the limit exists.}$$

Suggested practice questions:

1. Examples 18 (Parts (1), (3) and (4)) and 19 of Section 7.8 in the lecture outline.
2. Related questions from Section 7.8 in Homework 4.

Some Applications of Integrals.

- (1) Area between curves
- (2) Volume by disk/washer method
- (3) Volume by cylindrical shell method
- (4) Average value of a function
- (5) Arc length
- (6) Area of a surface of revolution

Suggested practice questions:

- 1. Examples 1 - 10 of Sections 6.1 - 6.3 in the lecture outline.
- 2. Examples 13 - 15 of Section 6.5 in the lecture outline.
- 3. Examples 1 - 5 of Sections 8.1 and 8.2 in the lecture outline.
- 4. Related questions from Sections 6.1 - 6.3, 6.5, 8.1 and 8.2 in Homework 5 and 6.

Convergence and Divergence of Sequences

1. Use functions to find limits of sequences
2. Some basic properties of convergent sequences
3. The Squeeze Theorem
4. A method to determine the divergence of a sequence

(see Fact 3 in Section 11.1 in the lecture outline)

Suggested practice questions:

1. Examples 1 and 2 of Section 11.1 in the lecture outline.
2. Related questions from Section 11.1 in Homework 7.

Convergence and Divergence of Series

1. Definition of convergence and divergence
2. Geometric series and telescoping series
3. Test for Divergence
4. Alternating Series Test
5. Definition of absolute convergence and conditional convergence

Suggested practice questions:

1. Examples 1, 2, 4, 5 and 6 of Section 11.2 in the lecture outline.
2. Examples 3 and 4 of Section 11.5 in the lecture outline.
3. Related questions from Sections 11.2 and 11.5 in Homework 7 and 8.

Tests for Positive Series, for Absolute Convergence and for Divergence

1. Integral Test and p -series
2. Direct Comparison Test and Limit Comparison Test
3. Ratio Test and Root Test

Suggested practice questions:

1. Examples 1 and 2 of Section 11.3 in the lecture outline.
2. Examples 1 and 2 of Section 11.4 in the lecture outline.
3. Examples 5 and 6 of Section 11.6 in the lecture outline.
4. Related questions from Sections 11.3, 11.4, 11.6 and 11.7 in Homework 7 and 8.

Power Series

1. Definition
2. Centre, radius of convergence, and interval of convergence

Suggested practice questions:

1. Examples 1 and 2 of Section 11.8 in the lecture outline.
2. Related questions from Section 11.8 in Homework 9.

Power Series Representations of Functions, Taylor Series and Maclaurin Series

1. The derivative and integral of a power series
2. Definition of Taylor series and Maclaurin series
3. Power series representation of a function obtained by Maclaurin series
4. The first six formulas on Maclaurin series from Table 1 in Section 11.10

Suggested practice questions:

1. Examples 1 - 3 of Section 11.9 in the lecture outline.
2. Examples 1 - 5 of Section 11.10 in the lecture outline.
3. Related questions from Sections 11.9 and 11.10 in Homework 9.